

Sparse Grids

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1.

Let S_n be the $(2^n + 1)$ -dimensional space of piecewise linear functions on an equidistant grid on the unit-interval with meshwidth $h_n = 2^{-n}$.

Let moreover T_n be the 2^{n-1} -dimensional subspace of S_n vanishing on all nodes (grid-points) of the mesh corresponding to S_{n-1} . Then the space S_n can be decomposed

$$(1.1) \quad S_n = S_o \oplus T_1 \oplus \dots \oplus T_n.$$

This decomposition is the standard hierarchical basis representation used e.g. in [1] if we use the standard local piecewise linear FE-basis in T_i , $i = 1, \dots, n$. Every such basis function is characterized by its support which is an interval $[(j-1)/2^{n-1}, j/2^{n-1}]$ for some $1 \leq j \leq 2^{n-1}$ or by its centerpoint $(j-0.5)/2^{n-1}$. The set of centerpoints we call the "grid" corresponding to an FE-space (see Fig. 1).

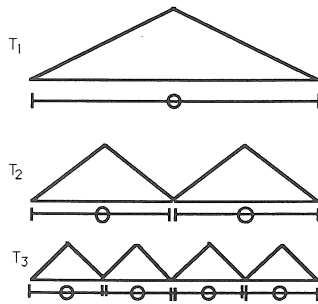


Fig. 1 Basis functions of T_1, T_2, T_3

Every $u \in S_n$ can be uniquely represented by the decomposition

$$(1.2) \quad u = u_o + \sum_{i=1}^n u_i \quad \text{with } u_o \in S_o, u_i \in T_i, i = 1, \dots, n$$

where every u_i can again be represented by a linear combination of the basis functions of T_i . Define now on the space $X = C^2([0, 1])$ the seminorm

$$(1.3) \quad |u| = \left\| \frac{\partial^2 u}{\partial x^2} \right\|_\infty \quad \text{for all } u \in X.$$

Let moreover u_I be the interpolant of u in S_n . Then we get from standard error analysis

$$(1.4) \quad \|u - u_I\|_\infty \leq \varepsilon = \frac{1}{2} \cdot 4^{-n-1} \cdot |u|.$$

If we use the decomposition (1.2)

$$u_I = u_o + \sum_{i=1}^n u_i \quad \text{with } u_o \in S_o, u_i \in T_i, i = 1, \dots, n$$

we get again by the same analysis using the ε defined in (1.4)

$$(1.5) \quad \|u_i\|_\infty \leq \frac{1}{2} 4^{-i} |u| = \varepsilon \cdot 4^{n-i+1}.$$

From this inequality we learn that the contributions of the subspaces T_i to the interpolant u_I tend to 0 in geometric progression if $i \rightarrow \infty$. This fact is the main reason for the usefulness of the hierarchical basis approach.

The inequality (1.5) can also be derived in a way which is easily generalizable to the 2-dimensional case in the next section. To this end we look at the basis representation of u .

Let for simplicity 0 be the center of the support interval of an arbitrary fixed basis function of T_i with length $2h_i$, $h_i = 2^{-i}$ and define the function

$$(1.6) \quad \begin{aligned} k_i(\xi) &= -(h_i - \xi)/2 \quad \text{for } \xi \geq 0 \\ &= -(h_i + \xi)/2 \quad \text{for } \xi < 0. \end{aligned}$$

Then we have

$$(1.7) \quad \begin{aligned} \int_{-h_i}^{+h_i} k_i(\xi) u''(\xi) d\xi &= \int_{-h_i}^{+h_i} -k_i'(\xi) u'(\xi) d\xi = \frac{1}{2} \int_{-h_i}^0 u'(\xi) d\xi - \frac{1}{2} \int_0^{h_i} u'(\xi) d\xi \\ &= u(0) - (u(h_i) + u(-h_i))/2 = c, \end{aligned}$$

where c is the coefficient of the basis function in the basis representation of u . On the other hand we have

$$(1.8) \quad \int_{-h_i}^{+h_i} k_i(\xi) u''(\xi) d\xi \leq |u| \cdot \int_{-h_i}^{+h_i} k_i(\xi) d\xi = \frac{h_i^2}{2} \cdot |u|$$

and with (1.7) we get (1.5).

Note that the bound (1.5) is equivalent to the proposition that for the basis representation of the function u the coefficients of the basis functions in the subspace T_i are bounded by the same constant. This leads to another interpretation of the hierarchical basis approach. To this end consider the sequence S_m for $m \rightarrow \infty$ and then remove from this sequence all subspaces T_i the contribution of which to u in the norm $\|\cdot\|_\infty$ is bounded by some prescribed bound ε . This just removes all T_i with $i > n$ for some n and we get the space S_n with some n depending on ε for all members of the sequence with index bigger than n . If we use this interpretation we can give a bound for the approximation error $\|u_I - u\|_\infty$ by

$$(1.9) \quad \|u_I - u\|_\infty \leq \sum_{i=n+1}^{\infty} \|u_i\|_\infty \leq \frac{1}{2} \sum_{i=n+1}^{\infty} 4^{-i} \cdot |u| = \frac{1}{2} \cdot \frac{4}{3} 4^{-n-1} \cdot |u|,$$

which is worse only by a factor $\frac{4}{3}$ (because of the use of the triangular inequality) than the sharp inequality (1.4).

This interpretation can also be adapted locally: It is not necessary to remove all basis functions of a subspace T_m but only those where the coefficients in the representation of u are small compared to ε . This approach is discussed in section 3. In the following section we extend the discussion to the 2-dimensional case.

2.

In the 2-dimensional case we consider rectangular meshes instead of triangular meshes. This is in contrast to the usual approach in the literature. We shall see that this choice based on cartesian products can be exploited in a significant reduction of the dimension of the FE-spaces. To this end let $S_{n,m}$ be the space of piecewise bilinear functions on an equidistant rectangular grid on the unit square with meshwidth $h_1 = 2^{-n}$ and $h_2 = 2^{-m}$ in x- resp. y-direction. To simplify the discussion we assume that all functions vanish on the boundary so that we do not need the 2-dimensional analogon of S_o in (1.1). Then

the dimension of $S_{n,m}$ is $(2^n - 1)(2^m - 1)$. Let moreover $T_{n,m}$ be the subspace of $S_{n,m}$ vanishing on all gridpoints corresponding to the grids of $S_{n-1,m}$ and $S_{n,m-1}$. Again we can introduce bilinear basis functions in $T_{n,m}$ which are characterized by their rectangular support $[(j-1)/2^{n-1}, j/2^{n-1}] \times [(k-1)/2^{m-1}, k/2^{m-1}]$ and we see immediately that $T_{n,m}$ is a 2^{n+m-2} dimensional space (cf. Fig. 2).

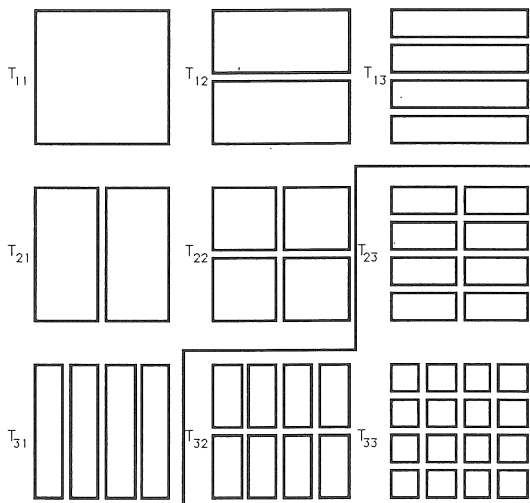


Fig. 2 The support of the basis functions in $T_{i,k}$.

The usual bilinear finite element space $S_{n,n}$ with rectangular elements can now be decomposed:

$$(2.1) \quad S_{n,n} = \sum_{i=1}^n \sum_{k=1}^n T_{i,k}$$

(note that here we use the fact that u vanishes on the boundary of the unit square). In contrast to the 1-dimensional case and also to the usual approach in two dimensions we have here a matrix scheme of subspaces instead of a linear chain.

Every $u \in S_{n,n}$ is correspondingly represented by

$$(2.2) \quad u = \sum_{i=1}^n \sum_{k=1}^n u_{ik} \quad \text{with} \quad u_{ik} \in T_{i,k}, \quad i, k = 1, \dots, n.$$

Define now the seminorm

$$(2.3) \quad |u| = \left\| \frac{\partial^4 u}{\partial x^2 \partial y^2} \right\|_{\infty}$$

defined on the space X of functions u where $\frac{\partial^4 u}{\partial x^2 \partial y^2}$ exists and is continuous, and let u_I be the interpolant of u in $S_{n,n}$ represented by

$$(2.4) \quad u_I = \sum_{i=1}^n \sum_{k=1}^n u_{ik} \quad \text{with} \quad u_{ik} \in T_{i,k}, \quad i, k = 1, \dots, n.$$

Similar to the 1-dimensional case we show that

$$(2.5) \quad \|u_{ij}\|_{\infty} \leq 4^{-i-j-1} |u|.$$

To this end consider the rectangular support of a basis function in $T_{i,j}$ and assume for simplicity that the origin of our coordinate system is at the center of this support.

Then we get with the functions defined in (1.6) by partial integration

$$\begin{aligned} & \int_{-h_i}^{+h_i} \int_{-h_j}^{+h_j} k_i(x) k_j(y) \frac{\partial^4 u(x,y)}{\partial x^2 \partial y^2} dx dy \\ &= u(0,0) - \frac{1}{2} (u(0, h_j) + u(0, -h_j) + u(h_i, 0) + u(-h_i, 0)) \\ & \quad + \frac{1}{4} (u(h_i, h_j) + u(-h_i, h_j) + u(h_i, -h_j) + u(-h_i, -h_j)) \\ &= c \end{aligned}$$

which is again the coefficient of the corresponding basis function in the basis representation of u_{ij} . On the other hand we have

$$\int_{-h_i}^{+h_i} \int_{-h_j}^{+h_j} k_i(x) k_j(y) \frac{\partial^4 u(x,y)}{\partial x^2 \partial y^2} dx dy \leq |u| \cdot \frac{h_i^2 h_j^2}{4}$$

from which (2.5) follows immediately.

Similar to the discussion at the end of section 1 we can also get a bound for the approximation error in terms of $|u|$ by considering the limit $n \rightarrow \infty$ and we get

$$\begin{aligned}
(2.6) \quad \|u - u_I\|_\infty &\leq \left\| \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} u_{ij} - \sum_{i=1}^n \sum_{j=1}^n u_{ij} \right\| \\
&\leq \sum_{i=1}^n \sum_{j=n+1}^{\infty} \|u_{ij}\| + \sum_{i=n+1}^{\infty} \sum_{j=1}^{\infty} \|u_{ij}\| \\
&\leq 2 \|u\| \sum_{i=1}^{\infty} \sum_{j=n+1}^{\infty} 4^{-i-j-1} \\
&= 2 \|u\| \sum_{i=1}^{\infty} 4^{-i} \sum_{j=n+1}^{\infty} 4^{-j-1} \\
&\leq \frac{2}{9} 4^{-n-1} \cdot \|u\| = \frac{1}{18} h_n^2 \cdot \|u\|.
\end{aligned}$$

We see that the order of convergence is $O(h^2)$ as it is natural for bilinear finite elements but the coefficient contains fourth derivatives instead of the second derivatives in standard error bounds.

If we analyse (2.6) we see that the use of a square scheme is disadvantageous. We would neglect the space $T_{1,n+1}$ with comparatively small dimension 2^n and rather large a contribution to the error of magnitude $4^{-n-3} \|u\|$ and we would include the subspace $T_{n,n}$ with relatively large dimension 2^{2n-2} and rather small a contribution to the error of magnitude $4^{-2n-1} \|u\|$. Following the strategy of section 1 we should better remove all subspaces with a contribution to the error smaller than some tolerance ε . Such a strategy leads to a triangular scheme of subspaces motivating the following definition of the approximation space

$$(2.7) \quad \hat{S}_{n,n} = \sum_{i=1}^n \sum_{j=1}^{n-i+1} T_{i,j}.$$

We have $\dim(\hat{S}_{n,n}) = (n-1)2^n + 1$ which is only $O(h_n^{-1} \log h_n^{-1})$ instead of $O(h_n^{-2})$ for $S_{n,n}$. In Fig. 2 the removed subspaces are indicated by a separating line. For the accuracy we obtain with

$$(2.8) \quad \hat{u}_I = \sum_{i=1}^n \sum_{j=1}^{n-i+1} u_{ij}, \quad u_{ij} \in T_{i,j}$$

$$\begin{aligned}
(2.9) \quad \| u - \hat{u}_I \|_\infty &\leq \left\| \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} u_{ij} - \sum_{i=1}^n \sum_{j=1}^{n-i+1} u_{ij} \right\|_\infty \\
&= \left\| \sum_{i=1}^n \sum_{j=n-i+2}^{\infty} u_{ij} + \sum_{i=n+1}^{\infty} \sum_{j=1}^{\infty} u_{ij} \right\|_\infty \\
&\leq \left(\sum_{i=1}^n \sum_{j=n-i+2}^{\infty} 4^{-i-j-1} + \sum_{i=n+1}^{\infty} \sum_{j=1}^{\infty} 4^{-i-j-1} \right) |u| \\
&= \left(n \cdot \frac{1}{3} 4^{-n-2} + \frac{4}{9} 4^{-n-2} \right) |u| \\
&= \left(\frac{1}{48} h_n^2 (\log h_n^{-1} + \frac{4}{3}) \right) |u|
\end{aligned}$$

which is only slightly worse (by a logarithmic factor) than the bound (2.6) for the approximation error in the space $S_{n,n}$.

As a result of this discussion we see that we have reduced the dimension significantly from $O(h^{-2})$ to $O(h^{-1} \log h^{-1})$ whereas the accuracy has been only slightly deteriorated from $O(h^2)$ to $O(h^2 \log h^{-1})$.

The deletion of certain subspaces implies the deletion of base functions and their corresponding gridpoints and results in a small subset of the original regular arrangement. Because of the analogy to sparse matrices we call such a grid a "sparse grid". For $n = 3$ and $n = 4$ such a grid is represented in Fig. 3. Every gridpoint represents an interpolation point i. e. the interpolant u_I coincides with u on all gridpoints.

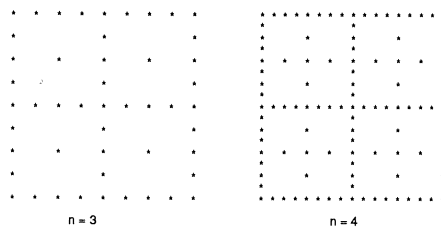


Fig. 3 Points in the square grid for $n=3$ and 4

3.

We restricted our attention here to functions vanishing on the boundary. If we drop this assumption we have in addition to represent 4 1-dimensional functions defined on the 4 boundary lines. These functions are represented as described in section 1 and raise the

dimension of the approximation space by $O(n)$. Thus this extension does not change the asymptotic rates given in this section.

In this section we say a few words about a data structure appropriate for storing functions of several variables using a hierarchical basis representation as described above. Let to every basis function g_i with support S_i correspond a node K_i in a graph G . Let moreover node K_i be connected with node K_j if and only if $S_i \supsetneq S_j$ and there exists no other basis function g_k with support S_k such that $S_i \supsetneq S_k \supsetneq S_j$. By this definition we obtain a directed acyclic graph G (a DAG). With each node we associate the coefficient of the basis function in the basis representation of u . By this construction in the 1-dimensional case every basis function in T_i is connected with exactly two basis functions in T_{i+1} and we get a binary tree with n levels. If we use the adaptive approach described at the end of section 1 we have to remove those leaves of the tree where the corresponding coefficient is smaller in modulus than some prescribed ε . This process has to be repeated as long as any leaves can be removed from the tree.

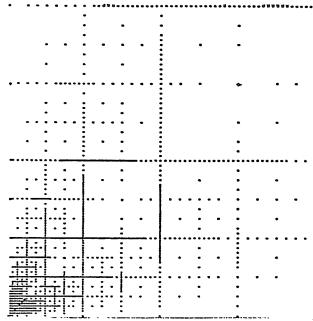


Fig. 4

In the 2-dimensional case a basis function of $T_{i,j}$ is connected with two basis functions of $T_{i,j+1}$ and two basis functions of $T_{i+1,j}$. In this case we don't get a tree because some basis function in $T_{i-1,j+1}$ is also connected to the basis functions in $T_{i,j+1}$ and we get only a DAG. Again we can apply the adaptive approach by removing on the lowest level the "leaves" in a repeated process. In Fig. 4 we show the grid pattern we get from this adaptive approach if we represent a singular function. We observe that in smooth regions we get a rather regular structure where the depth of the DAG is essentially constant. One can also apply this adaptive technique in a top down way. In this case a new level is added at all nodes where the corresponding coefficient is bigger than some prescribed ε . Of course one has to be very careful with this strategy because accidentally a coefficient might be very small though on lower levels the coefficients need not not be small. In the next section we shall use this top down technique for the solution of the Laplacian giving surprisingly good results.

4.

In this section we demonstrate that the FE-space introduced in section 2 and 3 is well suited for the solution of elliptic differential equations. As an example we solve the Laplacian

$$\Delta u = 0 \text{ on the unit square}$$

with Dirichlet boundary conditions

$$\begin{aligned} u(x, y) &= \sin \pi y \text{ if } x = 0 \\ &= 0 \text{ elsewhere on the boundary.} \end{aligned}$$

The exact solution is shown in Fig. 5 .

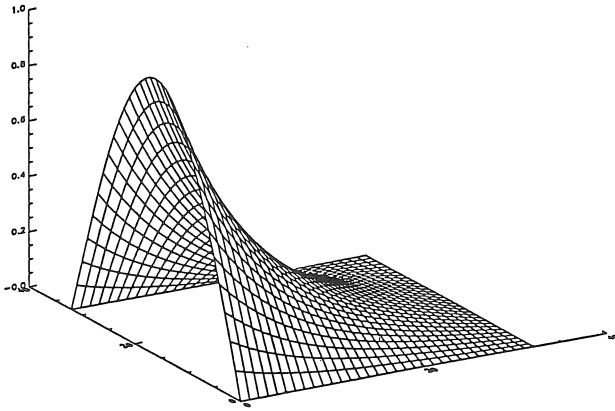


Fig. 5 Exact solution of the sample problem

It can be shown that we can perform one Gauss-Seidel iteration on the hierarchical basis with a constant number of operations per gridpoint. The algorithm uses a tree as datastructure and is formulated as a recursive procedure. Moreover a simple adaptive technique is easily implemented. If the coefficient of a certain basis function in the space $T_{i,k}$ is larger than some ε then the four new basis functions of the spaces $T_{i+1,k}$ and $T_{i,k+1}$ connected in the corresponding graph G are created and appended to the tree.

The grids and the error of the computed solution for $\varepsilon = 0.1, 0.01$ and 0.001 are shown in Fig. 6.

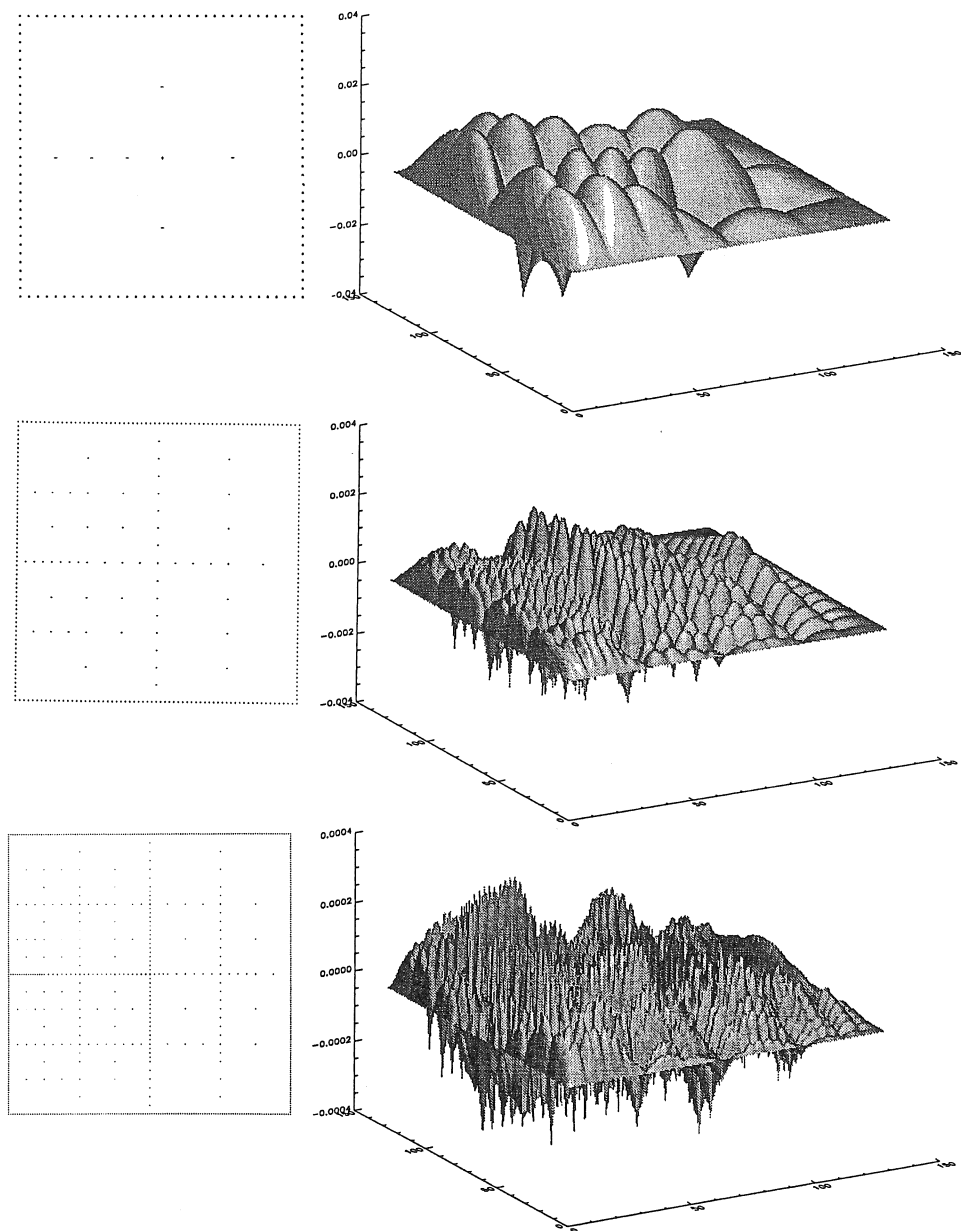


Fig. 6

We observe that for this example the error of the solution is bounded by the prescribed ϵ , i.e. the adaptive technique works satisfactorily in this example. The details of the code shall be explained in a separate paper.

Experimental results show that the convergence is as good as in other hierarchical basis codes. If this is true (we have no proof up to now) then the total work for the solution of the above equation is only proportional to h^{-1} times some power of $\log h^{-1}$ instead of h^{-2} . As we have already seen the storage is $O(h^{-1} \log h^{-1})$ instead of $O(h^{-2})$ whereas the accuracy is $O(h^2 \log h^{-1})$ instead of $O(h^2)$.

The economic storage requirement admits very small mesh sizes. On a 8MB workstation the smallest mesh size we used for the above example was $h = 2^{-14}$, a stepsize which is much smaller than the stepsize one could use with usual FE-spaces even on much bigger computers.

The same idea works even better in three dimensions. In this case the dimension of the usual FE-space is reduced from $O(h^{-3})$ to $O(h^{-1} (\log h^{-1})^2)$ whereas the interpolation error bound is $O(h^2 (\log h^{-1})^2)$ instead of the usual $O(h^2)$. Experiments worked satisfactorily for the Laplacian on the 3-dimensional unit cube and shall be reported in a separate paper.

Acknowledgement:

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641

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